MedCLIP-SAM v2

Problem

-Suboptimal Saliency Map Accuracy: researchers used gScoreCAM in v1 to generate the initial visual saliency maps from text prompts, however they found that this technique could be improved upon to increase zero-shot segmentation accuracy
-Inefficiency of Manual Text Prompts: v1 relied on simplistic keywords (for example brain tumor) provided by users which the researchers noted could limit the quality of the saliency maps
-Limited Modality Generalization: v1 was validated on only three modalities: breast ultrasound, brain MRI and lung X-ray

Dataset

-framework was validated across 4 diverse radiological modalities using the following datasets:
1-fine-tuning BiomedCLIP on MedPix & ROCO
2-Breast Ultrasound: BUSI & UDIAT datasets
3-Brain MRI: Brain Tumor Dataset (contains T1-weighted MRI scans for brain tumor segmentation)
4-Lung Chest: X-ray (COVID-QU-ExNormal, lung opacity, viral pneumonia, COVID-19 cases)
4-Lung CT: Konya 2020 (CT slices for fibrotic diseased lung segmentation)

Proposed approach: v2

1-Zero-Shot Pipeline:
1LLM-Guided Prompting: instead of simple keywords, the system uses GPT-4 to generate semantically rich clinical descriptions (shape, location, specific diagnostic features)
-M2IB Saliency Maps: the fine-tuned BiomedCLIP encoder uses the Multi-modal Information Bottleneck (M2IB) technique to align text and image features, this produces a continuous visual saliency map that filters out irrelevant information better than the previous version's gScoreCAM
-Coarse Mask Generation: the saliency map is binarized using Otsu's thresholding and refined via Connected Component Analysis (CCA) to remove noise and small artifacts
-SAM Refinement: this coarse mask provides bounding boxes or points to SAM (ViT-H) which generates the final refined zero-shot segmentation mask
2-Weakly Supervised Pipeline:
-system uses zero-shot masks as pseudo-labels to train standard seg network from scratch, it utilizes nnUNet framework to handle the training process
-Uncertainty Estimation: through checkpoint ensembling, the model saves multiple checkpoints during training cycles and averages their predictions to provide both a final robust mask and an entropy-based uncertainty map, this helps clinicians identify regions where the model is less certain, such as at tumor borders or artifacts

Evaluation & Results

1-Evaluation: researchers used 4 public datasets for validation: Breast Tumor Ultrasound, Brain Tumor MRI, Lung Chest X-ray, Lung CT
2-Validation Metrics: primary metrics used were the Dice-Sorensen Coefficient (DSC) and Normalized Surface Distance (NSD)
3-Comparison Methods: framework was benchmarked against SOTA zero-shot methods (SaLIP, SAMAug), few-shot methods (UniverSeg, ProtoSAM), the original MedCLIP-SAM v1
4-Obtained Results:
-Zero-Shot Segmentation: avg DSC increased from 64.54% (v1) to 77.61% (v2), avg NSD rose from 66.10% to 81.56%
-Modalities:
Breast Ultrasound: 77.76% DSC
Brain MRI: 76.52% DSC
Lung X-ray: 75.79% DSC
Lung CT: 80.38% DSC
-Weakly Supervised Segmentation: avg DSC improved from 70.90% (v1) to 82.11% (v2), avg NSD improved from 73.77% to 87.33%

Training ptotocol

1-BiomedCLIP Fine-Tuning: fine-tuning is performed on the MedPix dataset which contains over 20000 radiological image-text pairs, researchers use the Decoupled Hard Negative Noise Contrastive Estimation (DHN-NCE) loss which decouples positive and negative terms to allow for efficient training with small batch sizes and uses hard negative sampling to help the model distinguish between subtle, similar-looking medical features, the process fine-tunes the ViT and PubMedBERT encoders of the BiomedCLIP model
2-Zero-Shot Execution: medical image and its GPT-4 prompt are passed through the fine-tuned encoders, the M2IB module optimizes for mutual information between the image and text to locate the target region then the output is processed into visual prompts for SAM to generate the pseudo-masks
3-Weakly Supervised Training: framework trains nnUNet using the original medical images and the pseudo-labels from Stage 2, training runs for 600 epochs over 3 cycles, the last 10 checkpoints from each cycle (30 total) are saved, the predictions from all 30 checkpoints are averaged to produce the final probabilistic mask and the corresponding uncertainty estimation

Limitation

-Binary Segmentation Only
-Dimensionality Constraints: model is currently limited to 2D inputs

Future work

-Transition to 3D Medical Data
-Expanding Beyond Radiology: the team plans to test the framework on other types of biomedical imagery, including histopathological images and surgical videos
-Full Model Training: while the DHN-NCE loss was primarily used for unsupervised fine-tuning in this study, future work will test its effectiveness in training full models from scratch